6 A rectangle has length 8 cm
The rectangle has area 20 cm^2

The length of the rectangle is increased by 2 cm
The area of the rectangle is increased by 4 cm^2

Noah says,

"The width of the rectangle decreases by less than 5%"

Is Noah correct?
You must show how you get your answer.

Original width = 20 ÷ 8 = 2.5 cm

1 mark
Area ÷ length.

New width = 24 ÷ 10 = 2.4 cm

Decrease = 2.5 – 2.4 = 0.1 cm

Percentage decrease = 0.1 ÷ 2.5 × 100 = 4%

1 mark
Change ÷ original.

Yes, Noah is correct: 4% < 5%.

1 mark
4% < 5%.

(Total for Question 6 is 3 marks)